

Held-Out Cell Lines Are Not Held-Out Combinations: Auditing Evaluation Protocols for Drug Combination Synergy Prediction

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Computational prediction of drug-combination synergy is evaluated on benchmarks derived from combinatorial screens, in which a fixed panel of drug pairs is assayed across a fixed panel of cell lines. Recognising that uniformly random splits over measurements are permissive, the field has adopted cold-start protocols — leave-drug-out, leave-cell-line-out, leave-tissue-out. We show that holding out a cell line does not hold out a combination. Within a single screen the result is exact and follows from the screen’s own design: because a screen assays its pair matrix across its entire cell panel, leave-cell-line-out retains 100% of test combinations in training, on each of the four contributing screens we examine. Pooled over a harmonised DrugComb corpus of 180,746 feature-complete measurements the figure is 72.8%, pooling diluting the effect rather than reducing it. We quantify the consequence with a null model using no molecular information — a per-cell-line mean plus a per-pair mean. Under leave-cell-line-out this null model significantly outperforms both a symmetric neural baseline (+0.055 Pearson, $p = 0.0006$) and a 4.5M-parameter pathway-aware model (+0.122, $p = 0.0003$), while collapsing to 0.200 when combinations are held out. Using the pathway model as an instrument, we find that the sign of a small structure effect flips with protocol: structure helps where combinations are held out (+0.025, $p < 0.0001$) and hurts where cell lines are (−0.067, $p = 0.0018$), with two further protocols showing no resolvable difference. Component ablation attributes far more of the model’s advantage to imputation of missing drug–target annotations (−0.158 when removed) than to pathway topology (−0.017). The distinction that matters is between predicting a known combination in an uncharacterised context — legitimate and clinically relevant — and combination generalisation; one number cannot serve both. We release the audit, the null model, and fingerprinted split definitions so overlap can be reported alongside any cold-start result at negligible cost.

Keywords: Drug combination synergy; evaluation methodology; data leakage; benchmark design; null models.

1. Introduction

Combination therapy is standard in oncology, and the candidate space is far too large to screen exhaustively: for n compounds across m cell lines it grows as $O(n^2m)$. Computational prioritisation is therefore attractive, and a substantial literature has developed since DeepSynergy first applied deep learning to the problem.¹ Reviews count more than thirty machine-learning methods published since 2021 alone.^{2,3}

Progress is measured against benchmarks derived from combinatorial screens — O’Neil,⁴ NCI-ALMANAC,⁵ and their aggregation in DrugComb.^{6,7} These screens share a design property that is a virtue experimentally and a hazard statistically: a fixed panel of drug pairs is assayed across a fixed panel of cell lines. The resulting data is dense in *contexts per combination* and sparse in *distinct combinations*.

The field knows random splits are permissive here. Preuer et al. evaluated on novel combinations rather than random measurements as early as 2018, and reported that every method they compared degraded when extrapolating to unexplored drugs or cell lines.¹ Dewulf et al. formalised the cold-start subtasks for drug–drug interaction prediction and argued that matched validation schemes are critical to correct assessment.⁸ Contemporary methods routinely report leave-drug-out, leave-combination-out and leave-cell-line-out alongside headline numbers.^{9–11}

This is a specific instance of a general problem. Kapoor and Narayanan surveyed seventeen scientific fields and found leakage affecting 294 papers, sometimes producing wildly overoptimistic conclusions; in their reproducibility study of civil-war prediction, correcting leakage removed the entire advantage of complex models over decades-old logistic regression.^{12,13} Bennett et al. note that leakage is especially hard to detect in biological data because of complex dependencies between samples,¹⁴ and Rosenblatt et al. show that repeated subjects drastically inflate performance in connectome models¹⁵ — the direct structural analogue of repeated drug pairs.

We ask a question that, to our knowledge, has not been measured: **within each cold-start protocol, how much combination overlap actually survives?** Holding out a cell line removes a *context*; it does not remove a *combination*, because the same pair is measured in the cell lines that remain. Whether that matters is an empirical question about how redundant the screens are.

Contributions. (1) **A protocol audit** (Sec. 4.1). Across six protocols we measure the fraction of test combinations also present in training. Leave-cell-line-out retains 72.8% and leave-tissue-out 56.4%; within a single screen, leave-cell-line-out retains 100% by construction. (2) **A null model quantifying the consequence** (Sec. 4.2). A per-cell plus per-pair mean, using no molecular information, significantly outperforms two neural models under leave-cell-line-out. We argue it should be reported routinely. (3) **A demonstration that the protocol determines the verdict** (Sec. 4.3). On one architecture and corpus, the *sign* of a small structure effect flips with the protocol — positive where combinations are held out, negative where cell lines are, with two protocols showing no resolvable difference. (4) **A component attribution** (Sec. 4.4). The pathway model relies far more on imputing missing drug–target annotations (−0.158 Pearson when removed) than on pathway topology (−0.017), suggesting benefits attributed to biological priors may partly reflect database completion.

We are explicit about what we do *not* claim. We do not claim published results are invalid: we read the methods sections of three widely-cited methods and found their protocols sound (Sec. 2.3). We do not claim pathway representations are unhelpful in general; we test one architecture. The claim concerns what a class of protocols measures, and holds regardless of

how carefully any individual paper was executed.

2. Background and Related Work

2.1. *Synergy quantification*

Synergy is defined relative to a reference model of non-interaction, and the common references — Loewe additivity, Bliss independence, highest single agent, zero interaction potency — rest on different assumptions and disagree in practice.^{16–18} Vlot et al. found multiple instances of strong disagreement between the four on the Merck screen despite moderate overall concordance,¹⁶ and the choice of dose–response fitting software can flip a combination between synergistic and antagonistic.¹⁹ Implementations are available in SynergyFinder.²⁰ We use Loewe throughout and note in Sec. 6 that our overlap measurement is metric-independent while our performance results are not.

2.2. *Cold-start protocols*

Dewulf et al. identify the distinct subtasks arising when drugs, pairs or contexts are novel, and propose matched validation schemes.⁸ Our protocols follow this taxonomy; our contribution is measuring residual overlap *within* each scheme rather than proposing the schemes. Yousefabadi et al. state that permissive cross-validation allowing drug pairs across folds inflates performance, and adopt pair-held-out validation for antibacterial synergy.²¹ Abbasi and Rousu’s review concludes methods perform well for known drugs and cell lines but fall short for novel ones.² Agboola et al. report a random forest outperforming their deep model under cold-drug validation and interpret it transparently²² — a precedent for the null-model result we report.

2.3. *We verified, and do not allege, protocol failures*

Reading methods sections rather than abstracts: DeepSynergy’s headline 0.73 Pearson is for “novel drug combinations within the space of explored drugs and cell lines” — combination hold-out.¹ Monem et al. state their five-fold split ensures “each drug-drug combination is included in only one-fold”.⁹ SynProtX partitions randomly 80:20 for its headline and reports cold-start scenarios separately.¹⁰ Reported gains over DeepSynergy are, on this evidence, protocol-matched. The problem we identify is not that these papers erred; it is that a protocol they all use measures less than its name implies.

2.4. *Structure-informed synergy models*

Incorporating pathway or network structure is an active direction: network-proximity approaches,²³ pathway-activation scores,²⁴ interpretable subsystem models,²⁵ graph and transformer architectures,^{11,26,27} multi-task formulations,^{28,29} multi-omics ensembles,^{30,31} and proteomics-based models.¹⁰ Our instrument model belongs to this family; we use it to ask what such a model relies on, not to advance the family.

2.5. *Leakage in ML-based science*

Our finding instantiates a documented failure mode. Kapoor and Narayanan’s taxonomy of eight leakage types includes non-independence between train and test samples;^{12,13} REFORMS provides a reporting checklist;³² Bennett et al. give biology-specific guiding questions;¹⁴ Rosenblatt et al. quantify the effect in neuroimaging;¹⁵ and reviews in other applied fields document the same pattern.^{33,34} What is distinctive here is that the subfield already adopted a remedy — cold-start protocols — and the remedy is incomplete in a way that is invisible without counting pairs.

3. Methods

3.1. *Corpus*

We harmonise DrugComb⁶ to 634,631 measurements after removing self-pairs and rows lacking a Loewe score: 73,108 unique unordered pairs across 288 cell lines, a mean of 8.68 cell lines per pair (max 127). Requiring complete drug and cell features leaves 180,746 measurements over 1,945 drugs, 159 cell lines and 15 tissues, drawn principally from ALMANAC, FRIEDMAN, ONEIL and ASTRAZENECA.

Every protocol, split, overlap figure and performance number reported below is computed on this feature-complete subset of 180,746 measurements, not on the full 634,631; we state this explicitly because the distinction is precisely the kind this paper argues should be made. The filter preferentially retains well-characterised drugs, which appear in more pairs, so it plausibly biases the pooled overlap *upward*. Two things limit the concern: the per-screen replication in Sec. 4.1 reaches 100% by construction and so cannot be inflated by it, and `leave_combo_out` and `leave_drug_out` reach 0% regardless. We use the DrugComb v1.5 release.⁷

3.2. *Protocols*

Six protocols are implemented over the common corpus, each persisted as row indices with a corpus fingerprint so a split replays exactly without redistributing data: `random`, `leave_combo_out` (unseen combination, both drugs seen), `leave_drug_out` (one/both variants), `leave_cell_out`, `leave_tissue_out`, `leave_study_out`.

Grouped protocols assign entities by mass-aware greedy packing rather than uniform partitioning. The drug degree distribution is heavy-tailed and naive partitioning is unstable — an early implementation produced 297,039 test rows against 11,935 training rows. The packer orders groups by row mass, assigns each to the bin with the largest remaining deficit, and asserts the achieved test fraction lies in $[0.05, 0.25]$ before a split is accepted.

That assertion applies to the mass-packed protocols only. `leave_tissue_out` and `leave_study_out` partition on an explicit categorical assignment, so their test fraction is whatever the held-out block happens to weigh and the check is deliberately waived; we note this because a single held-out tissue can exceed the band substantially. `leave_tissue_out` is restricted to the eight tissues with at least 3,500 covered rows, the distribution being severely skewed (skin 31.4% of covered rows; pancreas 6 rows). Prostate, listed as eligible in an earlier version of our code, in fact holds 3,293 rows and is excluded.

Overlap metric. For a split we report the fraction of test unordered pairs that also appear in training. This requires no model and no training; it is a property of the partition, computable in seconds.

3.3. *Baselines*

B1, the null model, predicts a per-cell-line mean plus a per-pair mean deviation, both estimated on training rows only, backing off to the global mean for unseen entities. It uses no structure, targets, pathways or omics. Its role is diagnostic: a benchmark on which B1 is competitive is one whose signal is largely recoverable from which pair and which context, without chemistry.

We also report **B2**, gradient boosting on PCA-reduced embeddings — a learned but non-neural comparator that sits between B1 and the neural models — and **B3**, a symmetric MLP over drug embeddings and cell features with no ontology, graph or target projection, the reference for “does structure help”.

3.4. *The instrument model*

Pathway vocabulary, gene membership, graph topology and mechanism taxonomy are a versioned specification rather than code, so no assumption about relevant biology is embedded in the architecture. We use MSigDB Hallmark (50 gene sets)³⁵ and aggregate a pathway–pathway graph from STRING v12 physical interactions at combined score at least 700³⁶ with no hand-specified edges; panel genes absent from every Hallmark set — much of the druggable kinome, since Hallmark sets are expression-response signatures — are placed by STRING guilt-by-association (19 of 20 automatically). Drug target profiles project onto pathways, compose across the pair over the graph, and read out through a synergy head. 4.5M parameters.

Cell context is 160 dimensions: 14 PROGENy activities,³⁷ 64 principal components of protein-coding expression, 64 of CRISPR gene effect over the ontology’s gene universe, and 18 genotype flags, all from the DepMap portal³⁸ (CRISPRGeneEffect.csv, OmicsExpressionTPMLogp1HumanProteinCodingGenes.csv and OmicsSomaticMutations.csv; 18,531 genes across 1,208 cell lines). Drug structure uses MolFormer embeddings.³⁹ Drug–target profiles come from ChEMBL⁴⁰ over a 72-gene panel; only 24.7% of trainable rows have both drugs annotated, so a target-imputation head predicts profiles from structure (5-fold leave-drug-out macro-AUPRC 0.219, versus 0.162 nearest-neighbour and 0.064 prevalence prior).

3.5. *Statistics and reproducibility*

Comparisons use 6 seeds (1–5, 42) for `random`, `leave_combo_out`, `leave_drug_out` and `leave_cell_out`, where each seed redraws *both* the split partition and the model initialisation, since which pairs land in test is itself a large source of spread.

`leave_tissue_out` and `leave_study_out` are **deterministic by construction**: their partitions are fixed by an explicit tissue or study assignment, so the three seeds we run vary only the model initialisation and the split is identical across them. Their variances are therefore

Resolution floor. Repeating an identical configuration — same seed, same split, same code not comparable to the other four protocols — and are systematically smaller — a paired standard path — reproduces to within 0.007 Pearson, reflecting non-deterministic GPU kernels. We deviation of 0.001 in Sec. 4.2 reflects a fixed partition, not an unusually stable estimate. We therefore treat unpaired differences below about 0.01 as unresolved and say so where relevant report them as single-partition results with $n = 3$ initialisations and exclude them from the (Sec. 4.4). Paired differences are considerably tighter, because both models absorb the same abstract’s headline claims. Model-versus-model comparisons are paired on the shared split run-level noise and tested with a paired t -test. Targets are winsorised at the 0.5% quantile fitted on training rows only, then standardised; errors are reported in raw Loewe units.

4. Results

4.1. *Combination overlap survives cold-start protocols*

Table 1 reports the fraction of test combinations also present in training. The result is specific rather than a blanket dismissal. `leave_study_out` retains only 9.8%, because separate screens use largely disjoint drug panels: partitioning by *data source* approximates combination novelty, while partitioning by *biological context* does not.

Table 1. Fraction of test combinations also present in training, on the partitions all reported results use. Mean $\pm$ s.d. over 6 seeds, except the two deterministic protocols (Sec. 3.5), which have a single partition. See Fig. 1A.

protocol	test rows	test pairs also in train	holds out combinations
random	18,076	94.0% $\pm$ 0.3%	no
leave_cell_out	18,076	72.8% $\pm$ 0.1%	no
leave_tissue_out	71,638	56.4% (fixed)	no
leave_study_out	15,279	9.8% (fixed)	mostly
leave_drug_out	31,728	0.0%	yes
leave_combo_out	18,074	0.0%	yes

Within a single screen the overlap is total. Each contributing study is an independent screen with its own panels and protocol; repeating the audit inside each is a stronger test than the pooled corpus (Table 2, Fig. 3). The pooled figure of 72.8% *understates* the problem — pooling dilutes overlap by mixing in pairs unique to other screens. On O’Neil, the benchmark DeepSynergy introduced and still the field’s most common, a predictor with no molecular information reaches 0.682 under a random split and 0.600 under leave-cell-line-out.

Table 2. Per-study replication. Overlap with B1 Pearson in parentheses.

study	rows	pairs	rows/pair	random	leave_cell_out	leave_combo_out
ALMANAC	109,897	2,992	36.7	100% (0.677)	100% (0.653)	0% (0.057)
FRIEDMAN	50,107	2,926	17.1	100% (0.380)	100% (0.400)	0% (0.100)
ONEIL	9,369	315	29.7	100% (0.682)	100% (0.600)	0% (0.135)
ASTRAZENECA	5,910	407	14.5	100% (0.421)	92% (0.298)	0% (0.323)

4.2. The overlap is large enough to invert model rankings

Protocols in Table 3 are ordered by overlap, and B1 broadly tracks it ($r = 0.96$, $p = 0.003$; Spearman $\rho = 0.81$, $p = 0.05$). We deliberately do not lean on this. It is a correlation over six protocols, not six independent observations of a mechanism, and it does not survive scrutiny as a causal statement: at essentially zero overlap B1 still ranges from 0.111 (`leave_study_out`) to 0.200 (`leave_combo_out`), a spread of 0.089 that exceeds the +0.055 leave-cell-line-out gap the ordering is meant to explain. Something besides combination overlap moves B1 by more than the effect in question. The load-bearing evidence in this paper is the per-screen 100% result of Sec. 4.1, which is exact and needs no model; the correlation is a supporting observation only.

Table 3. Pearson correlation on Loewe. Mean over 6 seeds (3 initialisations for the two deterministic protocols). B2 is gradient boosting on PCA-reduced embeddings. See Fig. 1B.

protocol	overlap	B1 (no chemistry)	B2 (GBM)	B3 (MLP)	pathway model
random	94.0%	0.588	0.630	0.745	0.744
leave_cell_out	72.8%	0.425	0.375	0.370	0.304
leave_tissue_out	56.4%	0.325	0.277	0.202	0.211
leave_study_out	9.8%	0.111	0.146	0.139	0.123
leave_drug_out	0.0%	0.125	0.216	0.245	0.224
leave_combo_out	0.0%	0.200	0.378	0.594	0.618

Under `random`, B1 recovers 79% of a neural model’s correlation using no molecular information. Its drop once combinations are held out is 0.388 ± 0.008 — a tight estimate of what the protocol, rather than the model, was contributing.

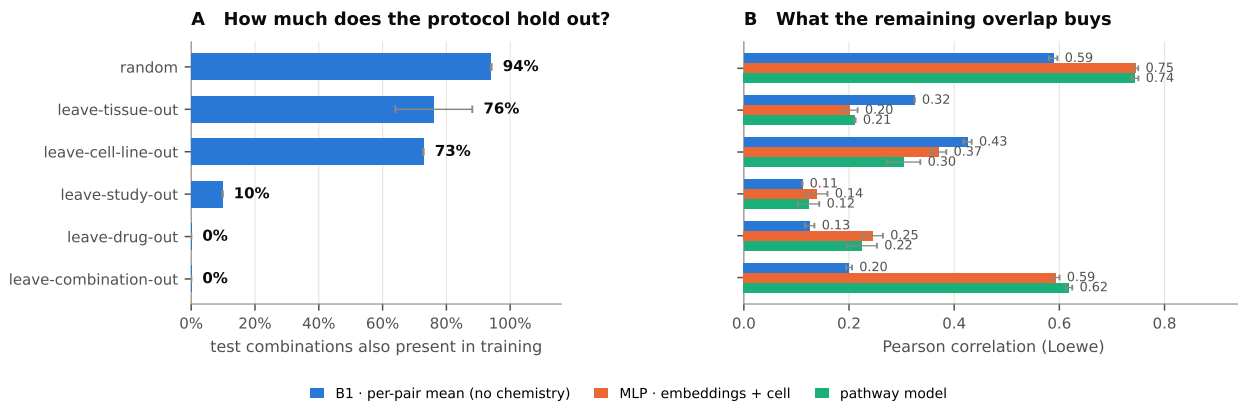
B1 beats both neural models on both high-overlap cold-start protocols, paired over seeds. Under leave-cell-line-out: B1 – MLP = +0.055 ($t = 7.71$, $df = 5$, $p = 0.0006$) and B1 – pathway model = +0.122 ($t = 8.81$, $df = 5$, $p = 0.0003$). Under leave-tissue-out, on a single fixed partition with three initialisations: B1 – MLP = +0.123 ($t = 11.85$, $df = 2$, $p = 0.007$) and B1 – pathway model = +0.113 ($t = 158.2$, $df = 2$, $p = 0.00004$). The very large t in the last comparison is an artefact of the fixed partition, not evidence of a stronger effect: B1 is a deterministic function of the split and so is constant across the three runs, leaving only the pathway model’s initialisation variance (paired s.d. 0.001) in the denominator. We report

it for completeness and rest the claim on the leave-cell-line-out arm, where the partition is redrawn.

The result is not peculiar to one hand-built null model. B2, an off-the-shelf gradient-boosting regressor, also exceeds both neural models under high overlap (Table 3) while losing to them by a wide margin (0.378 vs 0.594) once combinations are held out. We state the margins precisely rather than lean on the direction: under leave-cell-line-out B2 beats the pathway model comfortably (+0.071) but only matches the MLP (+0.005, inside the 0.007 resolution floor of Sec. 3.5). B2 is also a different function class, not the controlled comparison — an MLP handed the identifiable two-way structure — that would settle the question; we name that experiment in Sec. 6. What B2 does establish is that the effect is not an artefact of one tailored null model.

Eight paired tests are reported in this paper without correction for multiple comparisons. At a Bonferroni threshold of $\alpha = 0.05/8 = 0.00625$ all comparisons we describe as significant clear it except the leave-tissue-out B1 – MLP result ($p = 0.007$), which sits just outside and should be read as borderline.

Conversely, `leave_study_out` — the only protocol besides combination hold-out with low overlap — is hard for everything: B1 reaches 0.111 and the pathway model 0.123, both near the floor. A protocol that genuinely removes combination overlap produces uniformly low numbers, which is what a hard task should look like. This mirrors Kapoor and Narayanan’s civil-war result, where correcting leakage removed the advantage of complex models over logistic regression.¹²



Protocols ordered by combination overlap. Where overlap is high, a predictor with no molecular information of any kind (B1) is competitive with — under leave-cell-line-out, better than — neural models. Only when combinations are held out does the ordering become interpretable. Mean $\pm$ s.d. over seeds.

Fig. 1. (A) Combination overlap by protocol. (B) What that overlap buys each method. Protocols are ordered by overlap; the null model B1 tracks it at $r = 0.92$.

4.3. The protocol determines the verdict

Table 4 contains one small positive effect, one small negative effect, and two null results. We resist the temptation to call this “three incompatible conclusions”: the accurate and still-interesting statement is that *the sign of a small effect flips with the protocol*, on the same

two models, corpus, metric and target. The pattern is orderly — where combinations are held out, structure helps reliably if modestly; where they are not, it is level or harmful — and a single-protocol evaluation cannot distinguish the cases, which is the practical cost of the overlap in Sec. 4.1.

The effects are also small in absolute terms. The +0.025 and −0.067 here are an order of magnitude below the protocol-driven swings documented in Table 3, where the same pathway model ranges from 0.123 to 0.744 across protocols. Choice of protocol moves the number far more than choice of architecture does.

Table 4. Pathway model minus MLP. Both models see the identical split at each seed; paired t -test, $n = 6$.

protocol	Δ Pearson	t	p	conclusion supported
random	-0.001 ± 0.005	−0.58	0.59	structure is irrelevant
leave_combo_out	$+0.025 \pm 0.002$	29.7	< 0.0001	structure helps
leave_drug_out	-0.021 ± 0.036	−1.44	0.21	structure is irrelevant
leave_cell_out	-0.067 ± 0.027	−6.04	0.0018	structure hurts

4.4. What the pathway model actually relies on

Against the 0.007 resolution floor of Sec. 3.5, the four smallest rows of Table 5 are not resolved and we draw no conclusion from them. Removing the pathway *graph* — the component that makes the model pathway-aware — costs 0.017, while removing imputation of missing drug–target annotations costs 0.158.

Two cautions on how far this can be pushed. First, the ablations are single-seed, whereas the 0.007 floor was measured by repeating an *identical* configuration and therefore captures only run-to-run non-determinism, not variation across split draws and initialisations. It is the right floor for the −0.158 result, which exceeds it twentyfold, but it is too permissive to settle the −0.017 pathway-graph row, which we accordingly report as suggestive rather than established. Second, and more informative: removing the target branch *entirely* costs 0.157, statistically indistinguishable from removing imputation alone. The measured annotations contribute almost nothing on their own — only 24.7% of rows carry both — so what the target branch supplies is very nearly all supplied by the imputer.

Consistent with imputation carrying the model, performance stratified by annotation status inverts the naive expectation: pairs where *neither* drug is annotated reach 0.682, above pairs where *both* are (0.582). One reading is that a learned profile is denser and less noisy than a sparse measured one. It is not the only reading, and we flag the confound rather than resolve it: drugs are annotated in ChEMBL *because* they have been heavily studied, so annotation status also tracks drug class, assay history and literature attention. The two strata are not exchangeable, and this stratification should be read as consistent with the ablation rather than as independent evidence for it.

Table 5. Ablations on `leave_combo_out`. Single seed; see Fig. 2.

removed	PCC	Δ	resolved?
— (full model)	0.625	—	
graph-derived pair features	0.629	+0.004	no
drug-order symmetry	0.622	−0.002	no
projection mode (max→sum)	0.622	−0.003	no
observed-annotation mask	0.616	−0.008	no
pathway graph	0.607	− 0.017	yes
target-annotation imputation	0.467	− 0.158	yes
whole target branch	0.468	− 0.157	yes

We propose the reading that a meaningful part of what presents as “biological prior knowledge improves prediction” may be a model completing a sparse annotation matrix. That is useful, but it is a statement about database coverage rather than pathway biology, and invisible without component-wise ablation.

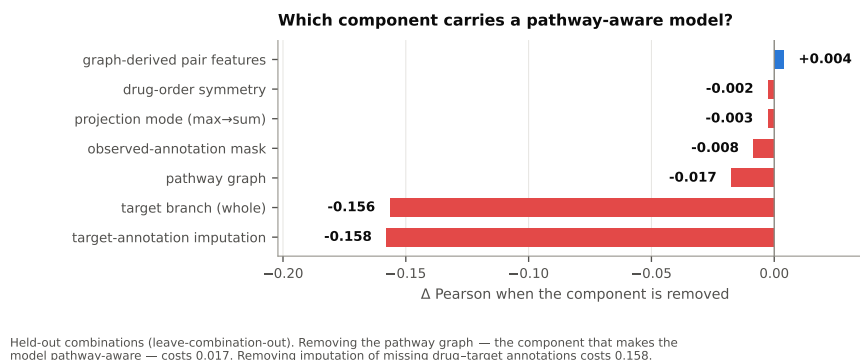
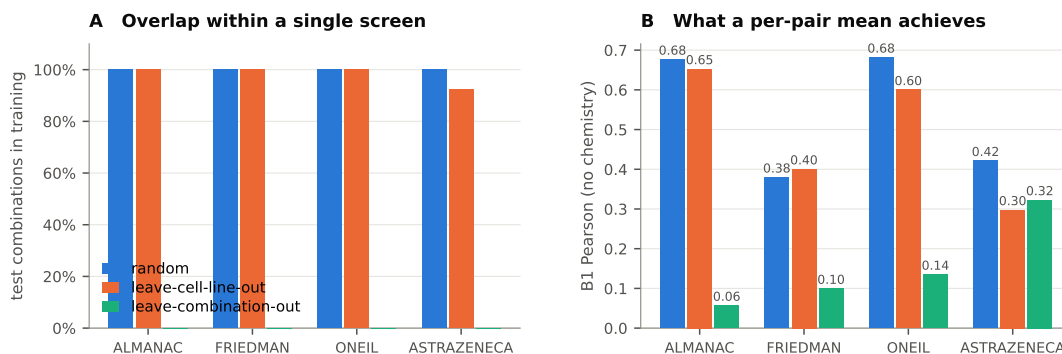


Fig. 2. Ablation deltas on `leave_combo_out`. Only the pathway graph and target-annotation imputation exceed the resolution floor, and they differ by an order of magnitude.

5. Discussion

Recommendations. Report combination overlap alongside any cold-start result: no model, no retraining, no new data — a property of the partition. Where the claim concerns combination discovery, prefer `leave_combo_out`; where it concerns external validity, prefer `leave_study_out` (9.8% residual overlap). Report B1, or any context-plus-pair null model, as a routine sanity check. Had it been reported, the leave-cell-line-out result would have been visible immediately. These are concrete instances of the reporting practices urged by REFORMS³² and by Bernett et al.’s guiding questions.¹⁴

What this does not say. Cold-cell-line prediction is not uninteresting. Predicting how a known combination behaves in an uncharacterised cell line is legitimate and clinically relevant



A combination screen assays its pair matrix across its whole cell panel, so within one screen holding out cell lines cannot hold out combinations. Overlap is 100% under both random and leave-cell-line-out; only leave-combination-out removes it.

Fig. 3. Per-study replication: within one screen the overlap is total. Each contributing study assays its pair matrix across its whole cell panel, so every held-out cell line still contains combinations seen in training.

— it is simply not a test of combination generalisation, and a model can excel at it while learning nothing about chemistry. The two should not be conflated, and one number cannot serve both.

Why this was not seen. The protocols are named for what they hold out, and the names are accurate: leave-cell-line-out does hold out cell lines. The gap is between the entity held out and the entity the task concerns. Because screens are dense in contexts and sparse in combinations, holding out the former barely touches the latter. This is invisible from model performance alone; it takes counting pairs.

Implications for structured biological priors. Sections 4.3 and 4.4 together counsel caution in attributing gains to biology. Where the pathway model helps, the effect is real but small, and ablation locates most of its advantage in annotation completion. Papers claiming benefits from biological priors would strengthen those claims by ablating knowledge-injection and data-completion pathways separately.

6. Limitations

One corpus. All results derive from DrugComb. Section 4.1 replicates the audit within four contributing screens, addressing pooling but not the harmonisation. **One instrument architecture.** Sections 4.3 and 4.4 rest on a single pathway model. We do not claim all such models behave this way; we claim the attribution is unmeasured in work reporting pathway benefits, and measuring it changes the interpretation. **No published model re-evaluated.** We quantify overlap and demonstrate consequences with our own baselines. Showing that specific published results change under an overlap-controlled protocol requires re-running those methods, and is the most important extension. The natural targets are MatchMaker²⁶ and the DrugCell subsystem model,²⁵ the latter being the canonical “biological structure helps” claim on which Sec. 4.4’s attribution most directly bears. **Baselines versus protocol.** Sec. 4.2

shows two non-neural baselines beating two neural ones under high overlap; it does not by itself establish that no neural model could recover the per-pair mean from embeddings. A hybrid — B3 given B1’s per-cell and per-pair terms as features — would separate “the protocol leaves too much signal” from “these particular networks failed to learn an identifiable two-way effect”, and we regard it as the first experiment a follow-up should run. **Loewe regression only.** Reference models disagree materially,^{16,17,19} so classification framings and other metrics may behave differently. The overlap measurement is metric-independent; the B1 consequences are not. **Resolution floor.** Unpaired differences below about 0.007 Pearson are not resolved (Sec. 3.5); four ablation rows fall below it and are reported as such.

7. Conclusion

Cold-start protocols in drug-combination synergy prediction hold out the entity they name, but not always the entity the task concerns. Leave-cell-line-out and leave-tissue-out retain most test combinations in training — all of them within a single screen — and on that protocol a predictor with no molecular information significantly outperforms neural models. Because the protocol determines whether biological structure appears to help, be irrelevant, or hurt, we recommend combination overlap and a context-plus-pair null model be reported as standard. Both are essentially free, and both would have surfaced this.

Data and Code Availability

Splits are persisted as row indices with a corpus fingerprint, reproducible without redistributing the corpus. Ontology specifications carry a content hash embedded in every checkpoint. The audit and null model run without a GPU in minutes. Code and split definitions will be made publicly available upon publication.

Use of Large Language Models

A large language model was used as a coding and drafting assistant: for parts of the analysis and plotting code, for LaTeX preparation, and for copy-editing of author-written text. All experiments, numerical results, and scientific claims were produced and verified by the authors, who take full responsibility for the content.

Acknowledgments

[TODO: funding sources and acknowledgments]

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